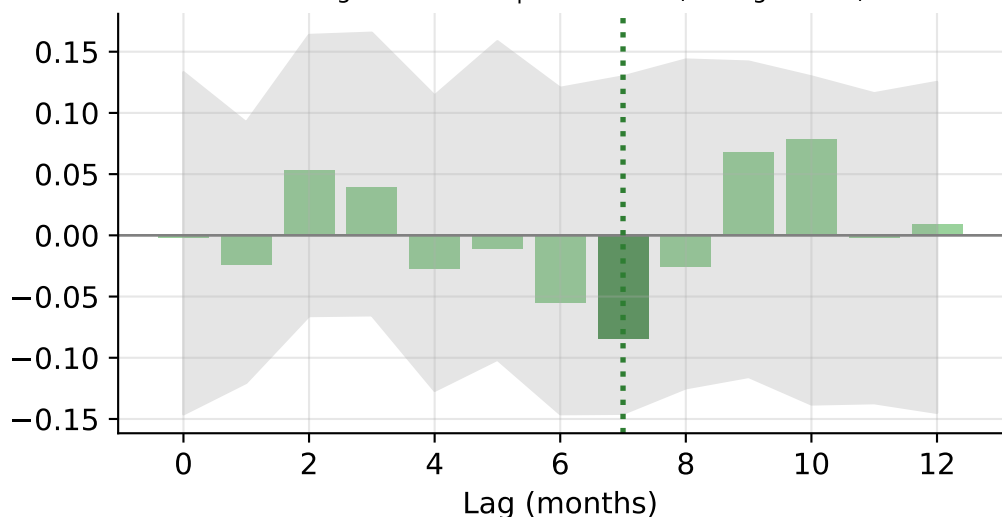
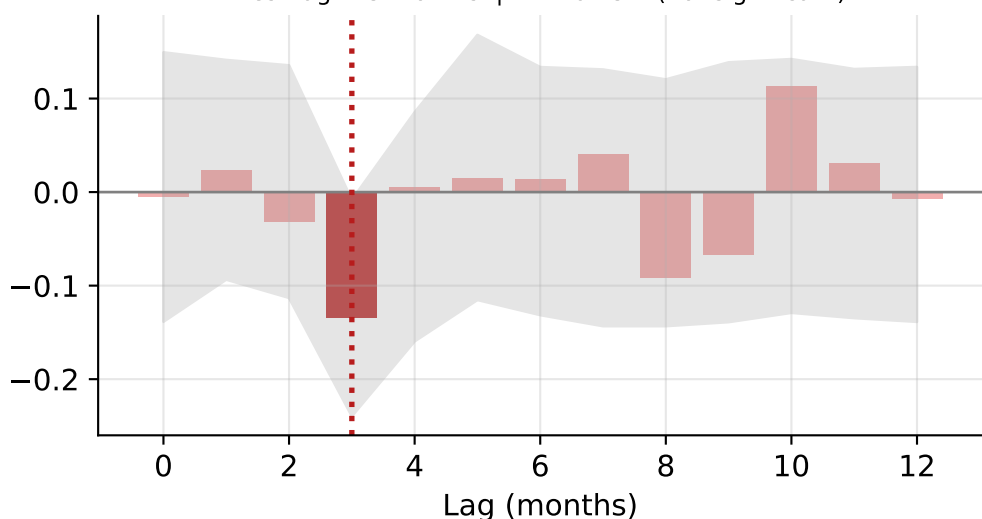


Figure 5 - Three-Step Causal Chain: Cross-Correlation at Each Path
Gray bands = 95% bootstrap CI | Dashed verticals = best lag

Step 1: GRACE TWS → MODIS Cropland NDVI
Best lag = 7 months | $r = -0.084$ (not significant)



Step 2: MODIS Cropland NDVI → Soybean Futures Log-Return
Best lag = 3 months | $r = -0.134$ (not significant)



Direct: GRACE TWS → Soybean Futures Log-Return
Best lag = 5 months | $r = -0.277$ ($p < 0.001$)

